Observation Theory for Estimation and Control: Consumer-Induced Geometry as a Complement to Observability, Filtering, and Feedback

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Abstract

Classical control theory separates several questions that are easy to conflate: whether a plant can be driven, whether its state can be inferred, how uncertainty should be filtered, and how actions should be chosen. A further question is usually supplied externally by the designer: *which distinctions in the estimated state matter to the downstream consumer?* Observation Theory (OT) supplies a candidate mathematical object for that missing interface. If a consumer $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ is judged in a local output metric G, its pullback $P_{\mathcal{C}}(x) = J_{\mathcal{C}}(x)^\top G(\mathcal{C}(x))\, J_{\mathcal{C}}(x)$ induces the local state geometry the consumer can distinguish, and estimator uncertainty Σ and consumer geometry compose as the operational uncertainty $U_{\mathcal{C}} = \mathrm{tr}(P_{\mathcal{C}}\Sigma)$. This paper defines a research campaign around that composition, organized as three interfaces to the classical stack: *estimator geometry* (a state-dependent $P_{\mathcal{C}}(x)$ moves the optimal estimate off the posterior mean), *resource value* ($V_{\mathcal{C}} = \mathrm{tr}(P_{\mathcal{C}}(\Sigma^- - \Sigma^+))$, the consumer-relative value of a measurement), and *dynamic relevance* ($S_{\mathcal{C}}(\Delta) = \mathrm{tr}[P_{\mathcal{C}}\, \Sigma(t+\Delta \mid I_t)]$, the age of operationally relevant uncertainty). We first establish a boundary: for a fixed positive-semidefinite $P_{\mathcal{C}}$ and an unconstrained Bayesian estimator, the posterior mean — hence the Kalman estimate — remains optimal; both this invariance and the rank-one value-of-observation identity are machine-checked in Lean 4/Mathlib (in their discrete and matrix forms) and have passed independent fresh-context derivation review. Each pillar of the program has close prior art in isolation: black-box recovery of quadratic task metrics (decision-focused learning), and consumer-derived weights composed with covariances (LQG sensing co-design, goal-oriented experimental design, value-of-information scheduling) in analytic form. The narrow novelty hypothesis is their conjunction: that a downstream consumer's operational metric can be recovered from the consumer itself under *query access only*, composed with classical estimator uncertainty, and validated prospectively, out of sample, at matched budgets that include the cost of probing. Evidence now exists for the full campaign: all seven experiments were sealed under the program's preregistration protocol and run once each at governed seeds — six passing every gate (blind recovery → scheduling with its analytic positive control; the reconstruction-matched two-consumer flip in sensing; signal-aware operational triggering; a physical-model optical-link consumer probed through the belief-averaged operator; and the multi-consumer tax, measured with its predicted angular shape), and the closed-loop campaign promoted on its third seal after two registered instrument misses whose corrections are part of the record (§12). The dynamic-relevance interface, pursued as its own track, has its first three sealed results as well: the age/staleness orderings cross at a point predictable in closed form, two consumers can disagree about which of two identically-aged histories is fresh, and a directional trigger beats the matched-budget isotropic signal-aware threshold by 16% — the honest baseline family, since signal-aware sampling already beats age-based scheduling. The conjunction has also met its first *neural* consumer: on a frozen, hosted, black-box language model consuming serialized numeric state, the read operator recovered by query-only logprob probing predicts the model's own held-out behavior, and precision allocated by it at matched serialization budgets beats task-blind allocation by 26% while capturing the full known-geometry advantage — with three sealed

instrument-and-design misses preceding the promotions on the public record. OT is proposed not as a replacement for observability, filtering, stability, or feedback, but as a complementary theory of operational distinguishability at their downstream boundary.

1 The missing question in the classical control stack

A modern estimation-and-control stack can be schematized as

$$\text{plant} \;\to\; \text{sensors} \;\to\; \text{estimator} \;\to\; \text{consumer(s)} \;\to\; \text{actions.}$$

Classical systems theory provides precise concepts for controllability and observability. Kalman filtering (Kalman, 1960) provides an optimal state estimate and error covariance under its linear-Gaussian assumptions. LQR, LQG, MPC, and stochastic control determine actions once an objective has been specified. Yet a design choice remains embedded in these methods: *which state errors actually matter?* In LQR this appears as the choice of a state penalty Q; in estimation it appears in accuracy specifications, sensor priorities, covariance weights, update rates, and communication policies. In many systems these choices are engineered successfully — but their operational geometry is externally specified.

Observation Theory begins one step downstream. Let $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ be a consumer and let D_Y be a twice-differentiable divergence on its output. Define the output metric and consumer Jacobian

$$G(z) \;=\; \nabla_v^2 D_Y(z, v)\big|_{v=z}, \qquad J_{\mathcal{C}}(x) \;=\; \frac{\partial \mathcal{C}}{\partial x}(x), \tag{1}$$

and the *read operator*

$$\boxed{P_{\mathcal{C}}(x) \;=\; J_{\mathcal{C}}(x)^\top \, G(\mathcal{C}(x)) \, J_{\mathcal{C}}(x).} \tag{2}$$

For sufficiently small state error $\delta = x - \hat{x}$,

$$D_Y\big(\mathcal{C}(x), \mathcal{C}(\hat{x})\big) \;\approx\; \tfrac{1}{2}\, \delta^\top P_{\mathcal{C}}(x)\, \delta, \qquad \mathbb{E}\, D_Y \;\approx\; \tfrac{1}{2}\, \mathrm{tr}\big(P_{\mathcal{C}}\, \Sigma_\delta\big), \tag{3}$$

so ordinary reconstruction error is the isotropic corner $P_{\mathcal{C}} = I$. The composition

$$U_{\mathcal{C}} \;=\; \mathrm{tr}\big(P_{\mathcal{C}}\, \Sigma\big) \tag{4}$$

reads: *what the estimator does not know*, weighted by *what the consumer can distinguish*. Neither object substitutes for the other: an accurate consumer model cannot estimate the plant; a perfect error covariance cannot say which residual uncertainty matters operationally.

The systems question is therefore not "can Observation Theory replace the Kalman filter?" It is: *given a classical estimator that tells us what we do not know, can a consumer-induced geometry tell us which parts of that uncertainty matter — and therefore where scarce sensing, communication, computation, and control resources should be spent?*

1.1 Three interfaces, not one

The program is organized around three distinct interfaces between the read operator and the classical stack:

$$P_{\mathcal{C}}(x) \;\longrightarrow\; \begin{cases} \textbf{Estimator geometry} & \text{if } P_{\mathcal{C}} \text{ varies with state (§3, Rem. 2),} \\ \textbf{Resource value} & \mathrm{tr}\big(P_{\mathcal{C}}\, \Delta\Sigma\big) \quad \text{(§4–§5),} \\ \textbf{Dynamic relevance} & \mathrm{tr}\big(P_{\mathcal{C}}\, \Sigma(\Delta)\big) \quad \text{(§9, conjectural).} \end{cases} \tag{5}$$

Interface 1 is *opened* rather than closed by the boundary theorem of §3: the posterior-mean invariance holds for fixed $P_{\mathcal{C}}$, and its failure under state-dependent $P_{\mathcal{C}}(x)$ is precisely where consumer geometry can change the estimator itself. Interface 2 is the paper's spine. Interface 3 is developed as an explicitly conjectural branch.

1.2 Access models: demonstrations, gradients, queries

The distinction that defines this program's niche is the access model. Inverse optimal control and inverse reinforcement learning recover objectives from *demonstrations*. Decision-focused learning differentiates through a *known* (or differentiable) consumer (Elmachtoub and Grigas, 2022; Amos and Kolter, 2017; Mandi et al., 2024). Observation Theory probes a *black-box* consumer under a declared observation distribution. Only the third access model is available when the consumer is a learned perception stack, a physical link, or another party's system — and those consumers, which resist hand-modeling, are exactly where a recovered geometry is necessary rather than convenient.

2 Observability is not operational distinguishability

Consider $x_{k+1} = Ax_k + Bu_k + w_k$, $y_k = Cx_k + v_k$. Classical observability asks whether distinct internal states can be distinguished from the measurement history. OT asks a downstream question: for consumer $\mathcal{C}$, two nearby states x and $x + \delta$ are *locally operationally distinguishable* according to $\delta^\top P_{\mathcal{C}}(x)\,\delta$. A direction in $\ker P_{\mathcal{C}}$ is locally invisible to that consumer; a direction with a large eigenvalue of $P_{\mathcal{C}}$ is strongly read. Hence a state direction can be perfectly observable to the sensor system and nearly irrelevant to the consumer; conversely, a very small estimation error can be operationally dominant if it lies in a strongly weighted read direction. These are complementary geometries on opposite sides of the estimator.

Two classical structures must be engaged here by name. First, *functional observability* (Darouach, 2000; Montanari et al., 2022) asks when a hand-specified linear functional $z = Lx$ can be reconstructed; OT's linear special case $\mathcal{C}(x) = Lx$ lies inside its subject matter, and OT adds the *metric* (not merely subspace) structure, nonlinear consumers, and recovery of L-like structure from the consumer itself. Second, for a linear consumer $z = Lx$ with output metric G, the pointwise OT geometry is $P_{\mathcal{C}} = L^\top GL$, and the output-weighted observability Gramian of frequency-weighted balanced truncation (Enns, 1984) (in its static-weight specialization) is its *dynamics-propagated accumulated analogue*,

$$W_{\mathcal{C}}(T) \;=\; \int_0^T e^{A^\top t}\, P_{\mathcal{C}}\, e^{At}\, dt. \tag{6}$$

Classical observability geometry thus *emerges from accumulated OT geometry* in the linear case: the Gramian is what the consumer's instantaneous read metric integrates to under the flow.

3 A boundary proposition: OT does not create a new Kalman filter

The first result of the campaign is deliberately negative.

Proposition 1 (Weighted posterior-mean invariance)**.** *Let X be an integrable random state with finite conditional second moment, Y the available observations, and $P \succeq 0$ a fixed state weighting. Then, over Y-measurable estimators $\hat{x}$, $\mathbb{E}\big[(X-\hat{x})^\top P\,(X-\hat{x}) \mid Y\big]$ is minimized over $\hat{x}$ by the posterior mean $\hat{x} = \mathbb{E}[X \mid Y]$.*

Proof. Let $m = \mathbb{E}[X \mid Y]$. Then $X - \hat{x} = (X - m) + (m - \hat{x})$, the cross terms vanish in conditional expectation, and

$$\mathbb{E}\big[(X - \hat{x})^\top P(X - \hat{x}) \mid Y\big] = \mathrm{tr}\big(P\,\Sigma_{X|Y}\big) + (m - \hat{x})^\top P\,(m - \hat{x}),$$

whose second term is nonnegative. □

The posterior mean already minimizes *every* fixed positive-semidefinite quadratic state loss; in the linear-Gaussian case this is the Kalman estimate. If the campaign merely rediscovered a weighted Kalman filter, it would not constitute the proposed contribution. The opportunity lies in deciding which measurements to take, which information to transmit, which state directions to retain, when to update, and eventually which operational errors to regulate — and, per Remark 2, in the state-dependent regime the invariance itself fails.

Remark 1 (Machine-checked form; weakened hypotheses). Proposition 1 (discrete form) and Proposition 2 below are machine-checked in Lean 4 against Mathlib (`lean/ObservationTheory/WeightedMeanInvariance.lean`; Lean and Mathlib v4.32.2, clean build, no `sorry`). The formalization shows the invariance survives under hypotheses weaker than stated: only $v^\top P v \geq 0$ for all v and weights summing to one are used — neither symmetry of P nor nonnegativity of the weights. We state the natural $P \succeq 0$ form above; the weakening is recorded here because the kernel checked it, not because the generality is operationally needed. The kernel check covers the discrete (finite-sample-space) forms; the continuous case rests on the standard argument. Per the program's charter, mechanical verification is necessary but not sufficient for a `[proved]` ledger classification, which additionally requires an independent fresh-context derivation pass; that pass has been run for both propositions (logged in the program ledger, VI-11/VI-12), confirming both and contributing two free strengthenings — only the symmetric part of P need be PSD, and P may be any Y-measurable (X-independent) weighting, sharpening the boundary against Remark 2.

Remark 2 (State dependence opens Interface 1)*.* Proposition 1 is for X-independent P. For a nonlinear consumer, $P_\mathcal{C}(x)$ depends on x; the loss $\mathbb{E}[(X-\hat{x})^\top P_\mathcal{C}(X)(X-\hat{x}) \mid Y]$ is then no longer a *fixed-weighting* quadratic, and its minimizer is in general not the posterior mean but the $P_\mathcal{C}$-tilted mean: writing P_s for the symmetric part of $P_\mathcal{C}$, the unconstrained minimizer is $\hat{x}^\star = \big(\mathbb{E}[P_s(X) \mid Y]\big)^{-1}\, \mathbb{E}[P_s(X)\, X \mid Y]$ whenever the belief-averaged weighting is invertible, and it equals the posterior mean iff $\mathbb{E}[P_s(X)(X-m) \mid Y] = 0$. Consumer geometry can thus change the estimator itself, not only the allocation of resources around it. The single-stage shift is elementary; what remains open in the campaign is its *recursive* form — the filtering analogue in which the tilted update propagates.

4 Consumer-relative value of observation

Let a Bayesian or Kalman estimator hold prior covariance Σ^- (symmetric) and consider a candidate measurement $y = Hx + v$, $v \sim \mathcal{N}(0, R)$, with $H\Sigma^- H^\top + R$ invertible (automatic for $R \succ 0$ and $\Sigma^- \succeq 0$). The covariance update gives $\Sigma^+ = \Sigma^- - \Sigma^- H^\top (H\Sigma^- H^\top + R)^{-1} H\Sigma^-$. Define the *consumer-relative value of observation*

$$V_\mathcal{C} \;=\; \operatorname{tr}\!\big(P_\mathcal{C}\,(\Sigma^- - \Sigma^+)\big) \;=\; \operatorname{tr}\!\big(P_\mathcal{C}\,\Sigma^- H^\top (H\Sigma^- H^\top + R)^{-1} H\Sigma^-\big). \tag{7}$$

Proposition 2 (Rank-one form)**.** *For symmetric Σ^-, a scalar measurement $y = h^\top x + v$ with $\operatorname{Var}(v) = r > 0$, and a rank-one consumer $P_\mathcal{C} = gg^\top$,*

$$V_\mathcal{C} \;=\; \frac{\big(g^\top \Sigma^- h\big)^2}{h^\top \Sigma^- h + r}. \tag{8}$$

(Symmetry of Σ^- is the load-bearing hypothesis — it is what collapses $h^\top \Sigma^- g$ into the square — while positive semidefiniteness is not needed for the identity itself; the Lean statement carries exactly $(\Sigma^-)^\top = \Sigma^-$ and the nonzero denominator, and the independent verification pass confirmed both the identity and the failure of the squared form for asymmetric Σ^-.)

This form is the section's message: a sensor is not operationally valuable because it observes a high-variance direction; it is valuable when its measurement direction h, *propagated through the current uncertainty* Σ^-, overlaps a direction g the consumer actually reads. The covariance algebra is classical. The proposed OT contribution is the *provenance* of g or $P_\mathcal{C}$: supplied by the downstream consumer's distinguishability geometry rather than chosen as an engineering weight.

5 Resource-constrained operation is the natural interface

Covariance objectives and their convex relaxations are established in sensor selection (Joshi and Boyd, 2009), and *weighted-trace* objectives in scheduling and selection (Dutta et al., 2023) and LQG sensing co-design (Tzoumas et al., 2021); the optimization $\min_a \operatorname{tr}(M\Sigma^+(a))$ for a designer-specified M is not an OT novelty. The proposed pipeline is

$$\text{consumer} \xrightarrow{\text{probe}} \widehat{P_\mathcal{C}} \xrightarrow{\text{compose}} \operatorname{tr}\big(\widehat{P_\mathcal{C}}\,\Sigma\big) \xrightarrow{\text{allocate}} a^\star,$$

with the allocation judged on the actual downstream consumer.

5.1 Probe cost is part of the objective

Recovering $\widehat{P_\mathcal{C}}$ — and re-recovering it along trajectories when the consumer is nonlinear — consumes exactly the resources OT allocates. The policy problem is therefore not $\min_\pi D_\mathcal{C}(\pi)$ subject to side constraints, but

$$\min_{\pi} \; D_\mathcal{C}(\pi) \;+\; \lambda_s \, C_{\text{sensing}}(\pi) \;+\; \lambda_c \, C_{\text{comm}}(\pi) \;+\; \lambda_p \, C_{\text{probing}}(\pi), \tag{9}$$

which contains a genuine exploration problem: *when is another consumer query worth its cost?* The marginal value of a probe — how much a further refinement of $\widehat{P_\mathcal{C}}$ changes downstream allocations — is itself an OT quantity. All campaign comparisons in §11 are run under (9), with probe cost on the same budget line as sensing and communication.

5.2 Greedy selection: a stated limitation

Weighted-trace objectives $\mathrm{tr}(P_\mathcal{C}\,\Delta\Sigma)$ are *not* supermodular in general: the counterexamples of Jawaid and Smith (2015) cover trace-of-covariance objectives, so greedy selection under (7) does not inherit the classical $(1-1/e)$ guarantee that log-determinant objectives enjoy. We do not attempt to repair this here. Available machinery includes approximate-supermodularity bounds (Chamon et al., 2022) and convex relaxations (Joshi and Boyd, 2009); establishing *conditions under which $V_\mathcal{C}$-greedy selection retains approximation guarantees* is registered as a campaign theorem — a named target, not a claimed result.

5.3 Operational event triggering

An operational trigger updates when the uncertainty visible to the consumer becomes excessive — rather than when a timer expires or isotropic variance crosses a threshold. The analytic special case is the value-of-information trigger of Soleymani et al. (2022; 2023), in which the transmit rule derived from the downstream LQG cost is provably optimal. Any OT triggering claim must cite that line as its analytic ancestor and either match it (LQG) or generalize beyond it (non-analytic consumers).

A registered finding from Campaign 4's calibration pilots (§12) sharpens the proposal: a *covariance-only* threshold $\mathrm{tr}(P_\mathcal{C}\,\Sigma_t) > \tau$ is direction-blind in effect — the covariance recursion is measurement-independent, so at a fixed budget every covariance functional collapses to a quasi-periodic schedule and the consumer weighting adds nothing. The operational trigger must be *signal- or belief-aware*: fire on the consumer-weighted *realized* discrepancy (e.g. $e^\top P_\mathcal{C}\, e > \tau$ for the innovation or sensor–remote gap e), consistent with the signal-aware designs of the VoI literature and with the belief-relative construction of §6.1.

6 Consumer geometry as a local control geometry

If the ultimate objective is expressed at a downstream output $z = \mathcal{C}(x)$ with desired value $z^\star = \mathcal{C}(x^\star)$, a second-order expansion gives $D_Y(\mathcal{C}(x), \mathcal{C}(x^\star)) \approx \frac{1}{2}(x - x^\star)^\top P_\mathcal{C}(x^\star)(x - x^\star)$, so $P_\mathcal{C}(x^\star)$ is a natural local state penalty — a candidate *provenance* for one component of an LQR/MPC objective, not a replacement for it. For a nonlinear consumer the penalty is trajectory dependent. Stability, robustness, terminal costs, actuator constraints, and feasibility remain control-theoretic problems; the proposed division of labor is that OT supplies where the state penalty comes from, and control theory determines what to do with it.

6.1 Threshold consumers and the belief-averaged read operator

The local metric (2) presumes a Hessian. Threshold-like consumers — bit-error rate, outage, any cliff — have degenerate or explosive G near the threshold, and the paper's own application exemplar (§14) is of this type. Where the local geometry exists, the robustified object is the *belief-averaged read operator*

$$\bar{P}_\mathcal{C}(b) \;=\; \mathbb{E}_{X\sim b}\big[\, J_\mathcal{C}(X)^\top\, G(\mathcal{C}(X))\, J_\mathcal{C}(X) \,\big], \tag{10}$$

where b is the estimator's belief: the read operator is averaged over the uncertainty the estimator actually has, rather than evaluated at a point the system may not occupy. For genuinely nonsmooth outage metrics no

Hessian exists to average, and a finite-perturbation or smoothed operational metric must replace (2) rather than pretending one does. *Belief-relative Observation Theory* — the read operator as a functional of the belief — is flagged as a substantial extension in its own right.

7 Black-box recovery is the sharp claim

If $P_{\mathcal{C}}$ must always be hand-specified, much of this program reduces to established task-weighted estimation and control. The stronger proposition is that the read geometry can, under stated conditions, be identified from the consumer itself: for appropriate consumers, finite-difference probes under a declared observation distribution recover sensitivity information without privileged access to the consumer's implementation. The probing *methodology* is itself prior art — empirical observability Gramians recover plant sensitivity by perturb-and-simulate (Krener and Ide, 2009) — and in machine learning, locally optimized decision losses recover quadratic task-metric surrogates from a black-box decision oracle by sampled perturbations (Shah et al., 2022; 2024). What is claimed here is the probed *object* (a downstream consumer's operational metric), its *composition* with an explicit estimator covariance, and the *validation protocol*: recovery on one perturbation distribution followed by success only on the same perturbations is insufficient; the test must be out of sample, at matched budgets, on the held-out consumer.

8 Multiple consumers

A real state service rarely has one downstream consumer. Given $P_1, \ldots, P_m$ for navigation, tracking, collision avoidance, communications, or mission logic, a weighted sum $P_{\mathcal{C}}^{\text{tot}} = \sum_i \lambda_i P_i$ can be useful but can hide incompatibility. Observation Theory's two-observer rate–distortion result supplies a precedent: in the Gaussian weighted-quadratic problem, successive refinement is lossless exactly when the observers' optimal residual uncertainty ellipsoids nest; distinct observers can demand geometrically incompatible information. That result is a coding theorem, not yet a control theorem. The corresponding systems question — when can one estimator, sensor schedule, or state representation efficiently serve several consumers, and when does their operational geometry force a multi-consumer tax? — was open when this campaign was designed and must not be answered by smuggling the coding result across. Campaign 6 has since measured it (§12): the tax exists, is exactly zero when the read geometries coincide, and rises monotonically with the principal angle between read subspaces.

9 Dynamic relevance: the age of operationally relevant uncertainty

Interface 3 is stated as an explicit conjecture. Let $\Sigma(t+\Delta \mid I_t)$ be the uncertainty predicted from currently retained information. Define

$$S_{\mathcal{C}}(\Delta) \;=\; \operatorname{tr}\big[\, P_{\mathcal{C}}(t+\Delta)\, \Sigma(t+\Delta \mid I_t) \,\big]. \tag{11}$$

An observation becomes stale when the uncertainty it leaves *in the consumer's read directions* becomes operationally expensive — age of information, in its signal-aware and semantics-aware refinements (Sun et al., 2020; Maatouk et al., 2023), weighted by operational geometry rather than by the clock. Our prior-art sweep found no anisotropic-staleness formulation; the nearest neighbor is the analytic value-of-information line (Soleymani et al., 2022), whose optimality results cover multi-dimensional Gauss–Markov processes. Because (11) has its own community, baselines (AoI, AoII, VoI), and falsifier, it is retained here as a conjectural branch and pursued independently as its own campaign track — whose first three sealed results now exist and are summarized in §12: the age/staleness orderings provably cross at a preregisterable point, two consumers can disagree about which of two identically-aged histories is fresh, and triggering on directional staleness beats the matched-budget isotropic signal-aware threshold (the falsifier in its honest form, since signal-aware sampling already beats age-based scheduling (Sun et al., 2020); no optimality claim attaches to either threshold heuristic). If long-horizon consequences dominate immediate consumer-relative covariance reduction, $S_{\mathcal{C}}$ may still remain a feature or terminal metric rather than a sufficient scheduling policy in richer settings; that outcome was a registered possible verdict of the sealed campaign (it did not occur in the synthetic class) and remains the standing boundary for the physical and random-delay settings ahead.

10 Prior art and the novelty hypothesis

The foundational claim must be narrower than several established areas, and — following a dedicated three-slice sweep (control theory; machine learning; communications and the application domain) — narrower still than the original draft of this paper stated. Each pillar of the hypothesis exists separately in strong form. Local quadratic task metrics are recovered from black-box decision oracles by sampled probing in decision-focused learning (Shah et al., 2022; 2024), and learned as Mahalanobis task metrics by end-to-end gradients through a differentiable task loss (Bansal et al., 2023). Consumer-derived weights composed with estimator covariances drive prospective sensor and bit allocation in LQG sensing co-design (Tzoumas et al., 2021), goal-oriented Bayesian experimental design (Attia et al., 2018; Wu et al., 2023; Neuberger et al., 2025) — including Jacobian/Hessian pullbacks of nonlinear goal maps with adjoint access — value-of-information scheduling (Soleymani et al., 2022; 2023; 2024), rate-cost control (Kostina and Hassibi, 2019; Tanaka et al., 2017), goal-oriented quantization (Zou et al., 2023), and task-based quantization (Shlezinger et al., 2019). Attention and anticipation select sensing resources against task-projected covariance metrics in closed loop (Carlone and Karaman, 2019); task-driven representations are co-designed by information bottlenecks (Pacelli and Majumdar, 2019); and consumer-derived perceptual metrics are established practice in vision (Zhang et al., 2018) and are consumed as training losses throughout learned compression. No novelty is claimed for any of these, nor for Kalman filtering, posterior-mean optimality, PSD-weighted covariance objectives, sensor selection and scheduling, active sensing, semantic communication, loss-calibrated inference (Lacoste-Julien et al., 2011), or inverse optimal control. The analytic LQG case is conceded entirely.

> **Research hypothesis (conjunction).** A downstream consumer induces an operational state geometry $P_{\mathcal{C}} = J^{\top} G J$ that can, under explicit observation conditions, be recovered from the consumer itself under *query access only* and a declared observation distribution; composed with classical estimator uncertainty as $\operatorname{tr}(P_{\mathcal{C}}\Sigma)$; and used to predict and optimize the operational value of sensing, communication, representation, and local control resources for consumers *outside the analytic families* — validated prospectively, out of sample, at matched budgets that include the cost of probing, on the actual downstream consumer.

This is a novelty hypothesis, not a novelty verdict: no single work found combines black-box probed consumer geometry, explicit covariance composition, and prospective matched-budget validation, but a referee holding decision-focused learning in one hand and goal-oriented experimental design in the other could reasonably call the conjunction an integration. The answer is the integration *plus the validation protocol* — reconstruction-matched flips, held-out consumers, matched budgets, physical endpoints — which no found work runs.

11 Research campaign

Campaign 1 — Verify the boundary

Random linear systems; direct optimization over admissible linear estimator gains compared with the Kalman solution under randomly drawn PSD consumer metrics. Any repeatable improvement over the Kalman estimate indicates an implementation error or a model violation. The invariance is classical and now machine-checked (Remark 1); the campaign *verifies* the boundary publicly rather than discovers it.

Campaign 2 — Consumer-relative value of observation

A dynamical state with heterogeneous candidate sensors and a planted low-rank consumer; selection by (7). The signature experiment is a *reconstruction-matched flip*: two sensing policies with essentially identical $\operatorname{tr}\Sigma$ but differently shaped uncertainty; only the consumer changes. Preregistered prediction: $\operatorname{tr}(P_{\mathcal{C}}\Sigma)$ identifies which policy is better for the downstream consumer while ordinary state MSE cannot. Where an analytic model exists, baselines must include the Riccati/VoI-derived weight — not merely hand-tuned or isotropic alternatives. *Status: sealed, run, all gates passed (§12).*

Campaign 3 — Blind recovery, then scheduling (flagship)

The point at which the pieces stop resembling analytic LQG/VoI machinery:

$$\text{unknown consumer} \;\rightarrow\; \text{query-only recovery of } \widehat{P_{\mathcal{C}}} \;\rightarrow\; \text{sensor scheduling} \;\rightarrow\; \text{held-out real consumer,}$$

at equal sensing budgets with probe cost included per (9), and with an *analytic-consumer positive-control arm*: on a consumer whose optimal weight is known (LQG/VoI), the blind pipeline is required to *recover and match* the known optimum — not beat it. Matching the analytic optimum blind is the calibration proof; winning on non-analytic consumers is the contribution. Success requires more than subspace recovery: the recovered operator must choose useful measurements and improve the held-out consumer at matched budget. *Status: sealed, run, all gates passed — including the positive control (§12).*

Campaign 4 — Operational event triggering

Limited transmissions or measurement energy; compare periodic, age-based, isotropic-covariance, hand-weighted, analytic-VoI (where available), and recovered-OT triggering. Primary outcome: downstream consumer performance at matched communication or energy — never lower OT surrogate loss by construction. *Status: sealed, run, all gates passed; its pilots contributed the covariance-null finding of §5.3 (§12).*

Campaign 5 — Physical consumer

A moving RF or free-space optical tracking system (§14). Navigation state includes position, velocity, attitude, angular rate, and structural motion; the physical consumer reads angle error, optical coupling, received power, BER, or outage. The key test is again reconstruction-matched: two navigation systems with similar state RMSE but different covariance geometry should produce different physical performance, and a blindly recovered $\bar{P}_{\mathcal{C}}$ should predict the ordering *before* the run. *Status: passed on the physics-based simulated consumer (§12); the hardware endpoint remains open.*

Campaign 6 — Multi-consumer state service

Two consumers with read subspaces ranging from aligned to orthogonal; compare common estimation, weighted scalarization, separate service, and layered/refined descriptions. The two-observer coding theorem supplies hypotheses about incompatibility but must not be imported as a control result; the goal is to determine whether a genuinely dynamic analogue of observer complementarity exists. *Status: sealed, run, all gates passed — the tax curve of §12.*

Campaign 7 — Closed-loop control

Only after the estimation and sensing claims survive: a recovered or derived $P_{\mathcal{C}}(x)$ inside an LQR/MPC objective against isotropic and competently hand-tuned baselines at matched control effort, with stability and constraint satisfaction independently audited, and the endpoint again the real downstream consumer. *Status: promoted on its third seal after two registered instrument misses — the record's proof of work (§12).*

12 First governed evidence

All seven campaigns have now been executed under the program's preregistration protocol: design registered before governed measurement; calibration pilots disclosed with every design change; effect floors frozen from calibration and sealed with harness code hashes; a *single* governed run per campaign at a sealed seed on fresh system draws; outcomes recorded in the program's public claims ledger regardless of sign, at class `[predicted]`. Table 1 summarizes; sealed preregistrations GO-P-2026-087/088/089 and PREREG-EC5/EC6/EC7-001 through EC7-003, harnesses, and committed result artifacts live in the program's campaign repository (EC track), with every gate re-derived from the committed per-system data by its verification script; the original sealing commits for 087–089 in the theory repository remain their binding registration timestamps, and Campaigns 5–7 are sealed natively in the campaign repository.[1]

Campaign 5 is the belief-averaged operator's first test on a threshold-like consumer: the link model (Gaussian far-field + fiber rolloff over a hidden state→physical map mixing attitude and range-scaled transverse position, operating at the Gaussian shoulder) is strictly query-access, its endpoint — coupling loss and outage when pointing with the estimate — is external link physics, and the smoothed finite-perturbation construction of §6.1 is what gets probed, exactly as that section prescribes for cliffs. Because the covariance path is deterministic, the E1 prediction (which trace-matched schedule serves the link better, read off $\mathrm{tr}(\bar{P}_{\mathcal{C}}\bar{\Sigma})$) *precedes* the loss measurement by construction — the paper's "predict the ordering before the run" test, passed at 94.1%. Scope, plainly: this is a physics-based *simulated* consumer — a step toward, not the arrival at, the hardware endpoint.

Campaign 6 answers the question §8 left open: a dynamic analogue of observer complementarity *exists*, as a measured tax curve. For planted consumer pairs whose read subspaces sit at a controlled principal angle (identical within-subspace weights, so the angle is the only contrast), the egalitarian price of serving both from one budget — best shared schedule over per-consumer utopia on the worst-of-two endpoint — is exactly 1.000 when the geometries coincide and rises monotonically to 1.149 at orthogonality. The two-observer coding theorem's nesting intuition motivated the design and was at no point imported as a result; the tax curve is the campaign's own measurement. One exploratory observation is reported without a gate: scalarized joint scheduling dominated time-sharing in 72 of 80 cells at this budget, so the successive-refinement analogue proper (layered descriptions) remains open.

Campaign 7 is the record's proof of work. Its three predictions — consumer-derived LQG penalty over isotropic at matched control effort, blind recover/match, and full geometry over the best endpoint-tuned diagonal — passed numerically in *all three* governed runs (K1 = 0.083/0.094/0.084 across seals, bars inherited unchanged throughout), yet the claims were promoted only on the third. The first seal failed its effort-matching integrity gate (absolute transfer, common-mode); the second failed the respecified cross-arm spread gate — arm-differentially, with the isotropic arms undershooting effort by ∼35% in one system, i.e. genuine comparison contamination that the gate existed to catch; both misses were recorded with no comparison claimed. The third seal replaced simulation-bisection matching with an *analytic* instrument (the Kalman recursion is control-independent, so expected effort follows exactly from joint second-moment propagation of $(x, \hat{x})$), whose measured cross-arm residual of 10^{-5} eliminated the error class; the held-out spread then read 0.041 against a 0.10 band, and the promotion followed. The methodological lesson generalizes: budget-matching instruments should be analytic wherever the dynamics permit, and an instrument's residual should be measured as its own diagnostic alongside the quantity it matches. The third seal's instrument has itself passed an independent fresh-context derivation audit: a from-scratch re-derivation in different coordinates $(x,\, x - \hat{x})$ agrees with the implementation to 10^{-15} relative on every audited configuration, and a 9×10^5-rollout falsification net bounds any residual bias below 0.03% — the arms were matched to the true closed-loop effort, not to a shared surrogate.

Three disclosures accompany the headline numbers, in the program's practice of reporting the record rather than the highlight reel. First, the pilots were load-bearing: Campaign 3's five calibration pilots caught transient-loaded probe charging, prior-inconsistent initial states, and noise-dominated policy comparisons (cured by common random numbers); Campaign 4's first pilot produced the covariance-null finding of §5.3 — a registered negative that redesigned the experiment; and Campaign 5's pilots respecified the anti-operator control to its load-bearing form (lose to the aligned schedule at equal probe charge) after a per-policy diagnostic showed the original all-comers form demanding the anti arm beat an unrelated weak agnostic policy — the same lesson Campaign 2 had already taught. Second, Campaign 2's governed invocation completed its measurement and then crashed in result serialization; the full output was committed as evidence, the serializer-only fix was recorded in a dated amendment, and the re-invocation was required to — and did — reproduce the first invocation's metrics line for line. Third, two of Campaign 2's gates passed exactly at their bars (the iso-control ceiling and the minimum-matched-systems count), which is noted in the ledger rather than hidden.

[1]Repository links are withheld for double-blind review. The quoted sealing-commit hashes and seal identifiers verify against the public program record; links are restored in the camera-ready.

Table 1: Governed OT-EC campaign results (single sealed runs; 20 fresh random LTI systems each; common random numbers across policies; probe cost charged where a recovered operator is used).

Campaign (prereg)	**Headline results (sealed gate)**	**Verdict**
C3 flagship: blind recovery → scheduling (087)	positive control: blind pipeline captures **94.6%** of the analytic optimum's gain (≥75%); wins **+16.3%** over the best consumer-agnostic policy on black-box consumers (≥8%); trace-matched ordering carried by $\mathrm{tr}(\widehat{P_{\mathcal{C}}}\Sigma)$ in **86.9%** of 61 pairs (≥65%)	ALL PASS 5/5
C2: the flip in sensing (088)	full two-consumer verdict inversion in **100%** of trace-matched systems (≥80%); $\mathrm{tr}(P_i\bar{\Sigma})$ predicts the ordering in **100%** of cells (≥85%); mean relative consumer gap **31.8%** (≥10%) at $\mathrm{tr}\,\bar{\Sigma}$ matched within 10%	ALL PASS 6/6
C4: operational triggering (089)	signal-aware $e^\top P_{\mathcal{C}} e$ trigger beats periodic/isotropic by **20.0%** (analytic arm) / **33.3%** (black-box arm) at matched realized budgets (≥10%); probe-charged blind trigger captures **86.9%** of the true-operator advantage (≥75%)	ALL PASS 6/6
C5: physical-model FSO consumer (EC5-001)	belief-averaged probe of a black-box optical-link model (§6.1): physical coupling-loss ordering of trace-matched schedules predicted *before the run* in **94.1%** of 17 pairs (≥80%); charged aligned schedule wins **+12.5%** coupling loss vs the best agnostic (≥4%); blind capture of an 8×-probe reference **89.8%** (≥70%)	ALL PASS 6/6
C6: the multi-consumer tax (EC6-001)	consumer pairs at controlled principal angles: the egalitarian best-shared-over-utopia tax runs **1.000/1.037/1.123/1.149** at 0°/30°/60°/90° — sharing exactly free when geometries coincide (≤1.05), a **14.9%** tax at orthogonality (≥1.05); consumer-aware sharing beats agnostic sharing by **12.5%** pooled on worst-of-two (≥5%)	ALL PASS 6/6
C7: closed-loop control (EC7-003, third seal)	consumer-derived LQG penalty at *analytically* matched control effort (instrument residual 10^{-5}): **+8.4%** vs the isotropic penalty on the consumer endpoint (≥4%); probe-charged blind penalty captures **99.5%** of the oracle advantage (≥60%); **+7.7%** vs the best endpoint-tuned diagonal (≥0); every loop stable	ALL PASS 8/8

Scope, stated plainly: these are synthetic LTI plants with planted or frozen consumers, single governed runs at class `[predicted]`, and greedy selection with no optimality guarantee (§5.2). What they establish is precisely the conjunction of §10 in its first instantiation — query-only recovery, covariance composition, prospective matched-budget validation with probe cost charged, and the recover/match positive control passed — and the two transfer results: the reconstruction-matched flip survives the move from codes to sensor schedules, and consumer weighting pays in the timing interface only through the realized error. The multi-consumer question is resolved through the tax curve and the closed-loop question through the third seal;

Table 2: Governed DR-track results (single sealed runs; 20 fresh random systems each; all preregistrations, harnesses, and result artifacts in the campaign repository).

Campaign (prereg)	**Headline results (sealed gate)**	**Verdict**
DR-1: the age/staleness crossover (DR1-001)	crossover predicted by closed form in **20/20** systems ($\geq$80%) and measured in **20/20** ($\geq$90%); $\lvert\theta^*_{\text{pred}} - \theta^*_{\text{meas}}\rvert$ median **0.67** grid steps (max 1.4 vs bar 4.0); the older observation wins below the crossing in all 20; isotropic-null flips 0/20	ALL PASS 7/7
DR-2: the staleness flip (DR2-001)	two-consumer verdict inversion on identical histories with identical age: predicted **19/20** ($\geq$80%; the 20th correctly predicted as no-flip and measured as none), measured **19/19** ($\geq$90%); pooled minimum relative gap **0.39** ($\geq$0.10)	ALL PASS 5/5
DR-3: scheduling at matched budgets (DR3-001)	directional trigger vs isotropic signal-aware threshold at matched realized budgets: **+16.3%** pooled consumer loss ($\geq$8%), positive 20/20; +65.3% over periodic ($\geq$30%, context); anti-control −61.5%; sealed angle structure Spearman −0.93 ($\leq$ −0.5); budget-integrity spread 0.58% ($\leq$2%)	ALL PASS 6/6

the hardware endpoint, richer consumer sets ($m > 2$, asymmetric priorities, layered descriptions), and the Interface-1 frontier (nonlinear consumers inside MPC, where $P_{\mathcal{C}}(x)$ moves the estimator itself) remain open.

The dynamic-relevance interface now has its own first evidence. Interface 3 (§9) was stated as an explicit conjecture and spun out as its own track (DR) in the campaign repository; its first three campaigns are sealed and measured (Table 2). The track's novelty discipline was set by its own anchor before any bar was frozen: signal-aware sampling already beats age-optimal sampling (Sun et al., 2020), so beating periodic or age-based scheduling is context, never the claim — only directional-beats-*isotropic-signal-aware* counts. DR-1 establishes the vocabulary: over a declared random system class, the age ordering and the directional-staleness ordering of two information sets provably disagree below a crossing angle computable in closed form before any rollout — a 6×-older observation of the slow mode beats a fresh observation of the fast mode wherever the consumer reads mostly slow — and the effect vanishes for the isotropic (trace) consumer. DR-2 transports the program's earlier coding-side (Campaign-2) flip from WHERE to WHEN: two consumers at canonical read angles declared before any system draw, given *identical* histories with *identical* age profiles, return opposite freshness verdicts; no consumer-blind freshness scalar can serve both. DR-3 is the operational claim: at matched realized transmit budgets, triggering on the realized error in the consumer's read direction beats the isotropic signal-aware threshold by **16.3%** pooled on the consumer endpoint (both are matched-budget heuristics; no optimality claim attaches), with the advantage carrying its sealed geometric signature — it shrinks (Spearman −0.93) exactly as the read direction rotates toward the noise-dominant mode, where the two triggers converge. The paper's pre-committed boundary outcome ($S_{\mathcal{C}}$ as a feature rather than a policy) was a registered possible verdict and did *not* occur in this system class. Scope, again plainly: smart-sensor idealization (exact-state transmission over a deterministic channel, so age-based and periodic coincide), planted known consumers, exact-linear plants; the probe-charged blind-recovery arm, random-delay channels (where the age-of-information line has its native force), and nonlinear plants are designated future seals.

The sharp claim now has a neural-consumer instantiation. The program's LM track sends the conjunction of §10 at a real language model: a frozen, hosted, black-box LLM (gemma-small on a managed research inference gateway) consuming a serialized 6-dim numeric state through planted question consumers, probed by query-only logprob finite differences with no gradient or attention access. Because a hosted endpoint is not bitwise-reproducible, the seal discipline adapts as it would for hardware: the governed artifact logs

Table 3: Governed LM-track results (single sealed runs per seal; a frozen hosted black-box LLM as the consumer; raw prompt/logprob logs are the committed artifacts).

Campaign (seals)	**Headline results (sealed gate)**	**Verdict**
LM-1: blind recovery from a hosted LLM (LM1-001/002/003)	probed $\widehat{P_{\mathcal{C}}}$ predicts the consumer's own held-out response: pooled Spearman **0.61** ($\geq$0.45, bars from two powered calibration draws), per-consumer 0.46/0.76 ($\geq$0.30); measured geometries separate at operator distance 0.41 ($\geq$0.25); stability 0.93/0.77 ($\geq$0.70); repeat/parse/transport instrument gates clean	promoted on the 3rd seal, ALL PASS 7/7 (two prior misses on the record: anchor class; calibration class)
LM-2: matched-budget precision allocation (LM2-001/002)	precision allocated by $\widehat{P_{\mathcal{C}}}$ at matched serialization budget: **+26.4%** pooled over the degeneracy-proof task-blind baseline ($\geq$12%), floors 0.13/0.40 ($\geq$0.05); blind capture of the oracle allocation **1.00** ($\geq$0.60); anti-aligned $-$38.6% ($\leq$+0.10); probing capital logged	promoted on the 2nd seal, ALL PASS 7/7 (one prior miss on the record: baseline degeneracy)

every prompt and returned logprob verbatim, so the entire gate evaluation is bitwise-reproducible from the committed record, with repeat-query agreement measured as its own instrument gate. Two campaigns are promoted (Table 3). LM-1 (recovery, third seal): the probed operator predicts the consumer's *own* response to held-out perturbation directions — pooled Spearman 0.61 against a bar calibrated from two independent powered calibration draws — with the two consumers' measured geometries separating at a preregistered operator distance. LM-2 (allocation, second seal): serialization precision allocated by the probed geometry at a *matched* budget beats a degeneracy-proof task-blind baseline by **26.4%** pooled on held-out consumer loss, with the blind allocation capturing the *full* oracle advantage (capture 1.00) — probing the black box is operationally equivalent to knowing the read components, at a logged probing capital cost. Three sealed misses precede the promotions and stay on the record: a task-oracle-anchored gate set (the consumer measurably deviates from the task ideal — gates must anchor to measured consumer behavior), a single-draw bar calibration (bars must come from the across-draw distribution at final power), and a baseline-degeneracy coincidence (a task-blind draw can coincide with the aligned allocation; the successor's mean-of-four distinct-subset baseline makes this impossible). One finding is reported ungated because it recurred without being sought: *coarsening the components the consumer does not read can actively improve the consumer* — the measured spurious-sensitivity channels operating in reverse — suggesting that context precision should be *deliberately* withheld outside the task's read geometry; the effect is consumer-dependent and is reported, never gated. Scope, plainly: one hosted model, planted numeric consumers, a fine/coarse serialization ladder; multi-model transfer, realistic task suites, and the staleness interface (LM-3) are designated future seals.

13 Falsifiers

The program narrows or fails if: the recovered read operator does not generalize outside its probing distribution; a conventional engineer-specified task matrix performs as well, making recovery operationally unnecessary; $P_{\mathcal{C}}(x)$ varies too rapidly for a local metric to hold over realistic estimator uncertainty (the belief-averaged operator (10) is the first line of defense, and its failure would bound belief-relative OT as well); probe cost under (9) consumes the advantage the recovered geometry provides; or long-horizon closed-loop consequences systematically defeat policies based on consumer-relative covariance reduction, in which case OT remains a theory of consumer-relative representation with local and terminal uses. These outcomes are scientific boundaries, to be reported, not patched away.

14 Application exemplar: cooperative optical high-altitude platforms

A useful application — not the foundation of the theory — is a constellation of persistent high-altitude platforms connected by free-space optical links (see (Kaymak et al., 2018) for mobile-FSO acquisition, tracking, and pointing generally). Each vehicle has a navigation and attitude state; each optical terminal reads only particular combinations of transverse displacement, line-of-sight angle, attitude, angular velocity, structural jitter, range, and optical state. Its physical consumer geometry need not resemble 6-DOF navigation MSE, and the standard practice in the pointing-acquisition-tracking literature is a static top-down pointing-error budget: the causal arrow runs forward (attitude error $\rightarrow$ link budget) only. The OT proposal runs it backward: probe the black-box receiver through modeled state perturbations, recover a belief-averaged read operator (10) — the endpoint metrics are threshold-like, so the smoothed construction of §6.1 is required, not optional — and contract it with the platform's Kalman covariance to decide when to request a beacon update, which navigation measurement to acquire, which information to transmit to a neighbor, and how much precision to allocate to each state direction. Multiple simultaneous links provide a natural multi-consumer test. The advantage of this exemplar is that its endpoint — received power, fiber coupling, BER, outage — is external and physical: OT cannot win merely by optimizing its own metric.

The prior-art posture here is stated in its narrowed form, following a targeted pass over the pointing-acquisition-tracking literature (SPIE proceedings included, inspected at abstract level where paywalled). Deriving sensor requirements from communication needs is established practice — canonically as a *static scalar* pointing-budget flow-down to star-tracker rate and gyro grade (Lee et al., 2005); probing the coupling surface by dither is likewise established — *for the fine-steering control loop* (Hu et al., 2020); and covariance-driven sensor scheduling exists as pure estimation theory with no link application (Al Ahdab et al., 2025). What no found work does is derive estimator design or scheduling from the link's *sensitivity geometry* (the Jacobian/Hessian of coupling or BER with respect to the navigation state) or compose a probed structure with the Kalman covariance; the three ingredients exist separately, and the composition appears open. The looser claim — that nobody derives estimator requirements from the link at all — is false and is not made.

15 Conclusion

Observation Theory proposes an estimation-and-control research program centered on a compact composition, $U_{\mathcal{C}} = \mathrm{tr}(P_{\mathcal{C}}\Sigma)$: the covariance belongs to estimation theory, the read operator to the consumer's operational geometry; the former says what is uncertain, the latter what matters. Three interfaces organize the program — estimator geometry when $P_{\mathcal{C}}$ varies with state, resource value through $\mathrm{tr}(P_{\mathcal{C}}\,\Delta\Sigma)$, and dynamic relevance through $S_{\mathcal{C}}(\Delta)$ — and none replaces observability, filtering, LQG, MPC, or stability theory. The campaign succeeds only if its sharpest claim survives: that $P_{\mathcal{C}}$ can be recovered from real downstream consumers under query access, at accounted probe cost, and prospectively predicts resource allocations that ordinary state reconstruction does not — matching the analytic optimum where one exists and winning where none does. All seven campaigns have now been run under seal — six passing outright, the closed-loop seventh promoted on its third seal after two registered instrument misses whose corrections are themselves part of the record (§12); positive controls included throughout. The dynamic-relevance track has followed with three sealed results of its own — the preregisterable age/staleness crossover, the two-consumer staleness flip, and the matched-budget scheduling win over the honest (isotropic signal-aware) baseline, with its predicted geometric signature. And the language-model track has carried the sharp claim to a real neural consumer: blind recovery that predicts the consumer's own held-out behavior, and matched-budget allocation that captures the full known-geometry advantage — five seals, two promotions, three named misses, all on the record. The hardware endpoint is where the claim now goes to be risked. The program is judged by reconstruction-matched flips, blind recovery, held-out dynamics, physical endpoints, competent classical baselines, and explicit negative results. Those gates have now been passed in the synthetic program, misses included on the record. If the physical endpoints fail, Observation Theory remains a theory of consumer-relative representation; if they pass, operational distinguishability becomes a working complement to the classical control stack.

Broader Impact Statement

This work is foundational estimation-and-control methodology evaluated entirely on synthetic systems and simulated physics under a sealed preregistration protocol. Its main societal-facing property is methodological: every claim is tied to a preregistered, hash-sealed, single governed run reported regardless of sign, with negative results and instrument failures on the public record — a practice we hope transfers. Consumer-aware sensing and scheduling could, like any resource-allocation machinery, be applied to surveillance systems; nothing in this paper is specific to such uses, and the operational gains demonstrated are for generic estimation consumers.

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